ASSOCIATION OF MATHEMATICS TEACHRS OF INDIA

Screening Test – BHASKARA Contest (NMTC Junior Level—IX and X Grades)

2025-2026

Instructions: (Read Carefully)

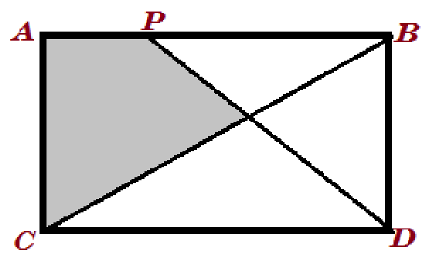
1. Fill in the Response sheet with your Name, Class and the Institution through which you appear, in the specified places.
2. Diagrams are only Visual guides; they are not drawn to scale.
3. You may use separate sheets to do rough work.
4. Use of Electronic gadgets such as Calculator, Mobile Phone or Computer is not permitted.
5. Duration of Test: 10 am to 12 Noon (Two hours)
6. For each correct response you get 1 mark; for each incorrect response, you lose ½ mark.

1. The greatest 4-digit number such that when divided by 16, 24 and 36 leaves 4 as remainder in each case is

A) 9994 **B)** 9940 **C)** 9094 **D)** 9904

2. *ABCD* is a rectangle whose length *AB* is 20 units and breadth is 10 units. Also, given $AP = 8$ units. The area of the shaded region is $\frac{p}{q}$ *sq unit*, where *p*, *q* are natural numbers with no common factors other than 1. The value of $p + q$ is

A) 167 **B)** 147 **C)** 157 **D)** 137



3. The solution of $\frac{\sqrt[7]{12+x}}{x} + \frac{\sqrt[7]{12+x}}{12} = \frac{64}{3}\left(\sqrt[7]{x}\right)$ is of the form $\frac{a}{b}$ where *a, b* are natural numbers with GCD $(a,b) = 1$; then $(b - a)$ is equal to

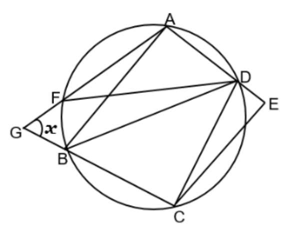
A) 115 **B)** 114 **C)** 113 **D)** 125

4. The value of $\left(52 + 6\sqrt{43}\right)^{3/2} - \left(52 - 6\sqrt{43}\right)^{3/2}$ is

A) 858 **B)** 918 **C)** 758 **D)** 828

5. In the adjoining figure $\angle DCE = 10^\circ$, $\angle CED = 98^\circ$, $\angle BDF = 28^\circ$. Then the measure of angle x is

A) 72° **B)** 76°
C) 44° **D)** 82°



6. *ABC* is a right triangle in which $\angle B = 90^\circ$. The inradius of the triangle is *r* and the circumradius of the triangle is R. If $R:r = 5:2$, then the value of $\cot^2\frac{A}{2} + \cot^2\frac{C}{2}$ is

A) $\frac{25}{4}$ **B)** 17 **C)** 13 **D)** 14

7. If (α , β) and (γ , β) are the roots of the simultaneous equations:

$$|x-1|+|y-5|=1; \qquad y=5+|x-1|$$

then the value of $\alpha+\beta+\gamma$ is

A) $\frac{15}{2}$ **B)** $\frac{17}{2}$ **C)** $\frac{14}{3}$ **D)** $\frac{19}{2}$

8. Three persons Ram, Ali and Peter were to be hired to paint a house. Ram and Ali can paint the whole house in 30 days, Ali and Peter in 40 days while Peter and Ram can do it in 60 days. If all of them were hired together, in how many days can they all three complete 50% of the work?

A) $24\frac{1}{3}$ **B)** $25\frac{1}{2}$ **C)** $26\frac{1}{3}$ **D)** $26\frac{2}{3}$

9. $\frac{\sqrt{a+3b}+\sqrt{a-3b}}{\sqrt{a+3b}-\sqrt{a-3b}}=x$, then the value of $\frac{3bx^2+3b}{ax}$ is

A) 1 **B)** **2** **C)** 3 **D)** 4

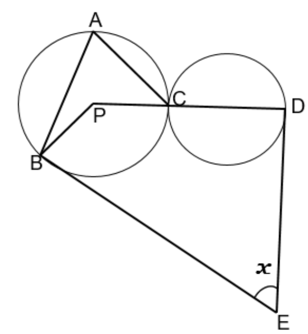
10. The number of integral solutions of the inequation $\left|\frac{2}{x-13}\right|>\frac{8}{9}$ is

A) 1 **B)** **2** **C)** 3 **D)** 4

11. In the adjoining figure, P is the centre of the first circle, which touches the other circle in C. PCD is along the diameter of the second circle.

∠PBA = 20° and ∠PCA = 30°.

The tangents at B and D meet at E. The measure of the angle x is



A) 75° **B)** 80° **C)** 70° **D)** 85°

12. If α , β are the values of x satisfying the equation $3\sqrt{log_2 x}-log_2 8x+1=0$, where α < β, then the value of $\left(\frac{\beta}{\alpha}\right)$ is

A) 2 **B)** 4 **C)** 6 **D)** 8

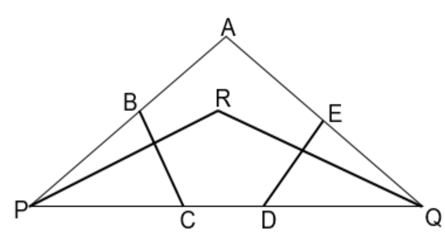
13. When a natural number is divided by 11, the remainder is 4. When the square of this number is divided by 11, the remainder is

A) 4 **B)** 5 **C)** 7 **D)** 9

14. The unit's digit of a 2-digit number is twice the ten's digit. When the number is multiplied by the sum of the digits the result is 144. For another 2-digit number, the ten's digit is twice the unit's digit and the product of the number with the sum of its digits is 567. Then the sum of the two 2-digit numbers is

A) 68 **B)** 86 **C)** 98 **D)** 87

15. *ABCDE* is a pentagon. ∠*AED* = 126°, ∠*BAE* = ∠*CDE* and ∠*ABC* is 4° less than ∠*BAE* and ∠*BCD* is 6° less than ∠*CDE*. *PR*, *QR* the bisectors of ∠*BPC*, ∠*EQD* respectively, meet at *R*. Points P, C, D, Q are collinear. Then measure of ∠*PRQ* is



A) 151° **B)** 137° **C)** 141° **D)** 143°

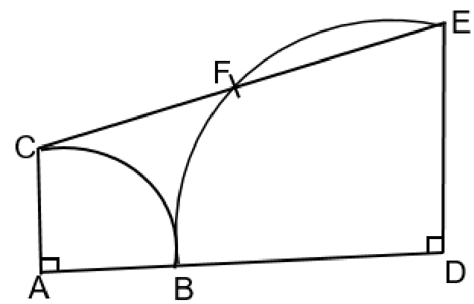
FILL IN THE BLANKS

16. a, b, c are real numbers such that $b - c = 8$ and $bc + a^2 + 16 = 0$.
The numerical value of $a^{2025} + b^{2025} + c^{2025}$ is ______.

17. Given $f(x) = \frac{2025x}{x+1}$ where $x \neq -1$. Then the value of x for which $f(f(x)) = (2025)^2$ is _______.

18. The sum of all the roots of the equation $\sqrt[3]{16 - x^3} = 4 - x$ is ________.

19. In the adjoining figure, two Quadrants are touching at B. CE is joined by a straight line, whose mid-point is F. The measure of $\angle CED$ is _______.



20. The value of k for which the equation $x^3 - 6x^2 + 11x + (6 - k) = 0$ has exactly three positive integer solutions is _______.

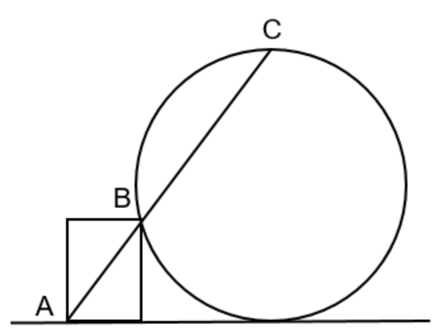
21. The number of 3-digit numbers of the form $ab5$ (where a, b are digits) which are divisible by 9 is _______.

22. If $a = \sqrt{(2025)^3 - (2023)^3}$, the value of $\sqrt{\frac{a^2-2}{6}}$ is _______.

23. In a math Olympiad examination, 12% of the students who appeared from a class did not solve any problem; 32% solved with some mistakes. The remaining 14 students solved the paper fully and correctly. The number of students in the class is ________.

24. When $a = 2025$, the numerical value of
$$|2a^3 - 3a^2 - 2a + 1| - |2a^3 - 3a^2 - 3a - 2025| \text{ is } \_\_\_\_\_\_\_.$$

25. A circular hoop and a rectangular frame are standing on the level ground as shown. The diagonal AB is extended to meet the circular hoop at the highest point C. If $AB = 18cm$, $BC = 32cm$, the radius of the hoop (in cm) is ________.



26. 'n' is a natural number. The number of 'n' for which $\frac{16(n^2-n-1)^2}{2n-1}$ is a natural number is ___.

27. The number of solutions (x, y) of the simultaneous equations
$log_4 x - log_2 y = 0, \quad x^2 = 8 + 2y^2$ is ______.

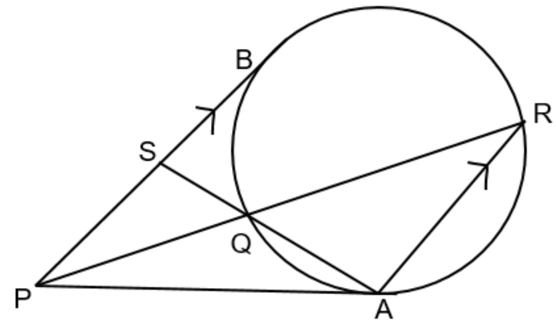
28. In the adjoining figure,

PA, PB are tangents.

AR is parallel to *PB.*

$PQ = 6$; $QR = 18$.

Length SB = _________.



29. A large watermelon weighs 20*kg* with 98% of its weight being water. It is left outside in the sunshine for some time. Some water evaporated and the water content in the watermelon is now 95% of its weight in water. The reduced weight in *kg* is _______.

30. In a geometric progression, the fourth term exceeds the third term by 24 and the sum of the second and third term is 6. Then, the sum of the second, third and fourth terms is ______.

End of Question Paper